



PDP, ICE & PFI – Installation Guide

This document provides an end-to-end checklist for installing, verifying, and using **Partial Dependence Plots (PDP)**, **Individual Conditional Expectation (ICE)**, and **Permutation Feature Importance (PFI)** across multiple environments:

- Jupyter Notebook
- Local Environment Terminal (Command Line)
- Google Colab

Key Note (Important for Learners)

- PDP, ICE, and PFI are techniques and they are primarily implemented using:
 - scikit-learn (sklearn.inspection)
 - *(Optional)* pdpbox for advanced PDP & ICE visualisations

1. Jupyter Notebook (pip-based)

Required Library

- **scikit-learn**

Install Required Library

```
!pip install scikit-learn
```

(Optional) Install PDPBox for Advanced PDP/ICE

```
!pip install pdpbox
```



Verify Installation

```
import sklearn
print(sklearn.__version__)
```

Import PDP & ICE (scikit-learn)

```
from sklearn.inspection import PartialDependenceDisplay
```

- Supports both **PDP** and **ICE** via `kind="average"` or `kind="individual"`

Import PFI (Permutation Feature Importance)

```
from sklearn.inspection import permutation_importance
```

2. Local Environment / Terminal (pip)

Install Required Library

```
pip install scikit-learn
```

(Optional) Install PDPBox

```
pip install pdpbox
```



Verify Installation

```
python -c "import sklearn; print(sklearn.__version__)"
```

Import PDP & ICE (scikit-learn)

```
from sklearn.inspection import PartialDependenceDisplay
```

- Supports both **PDP and ICE** via `kind="average"` or `kind="individual"`

Import PFI (Permutation Feature Importance)

```
from sklearn.inspection import permutation_importance
```

3. Conda Environment (Anaconda)

Scikit-learn is usually **pre-installed** in Conda environments.

Check Version

```
conda list scikit-learn
```

(Optional) Install PDPBox

```
pip install pdpbox
```

PDPBox is not officially available via conda



Import PDP & ICE (scikit-learn)

```
from sklearn.inspection import PartialDependenceDisplay
```

- Supports both **PDP and ICE** via `kind="average"` or `kind="individual"`

Import PFI (Permutation Feature Importance)

```
from sklearn.inspection import permutation_importance
```

4. Google Colab

Scikit-learn is **pre-installed** in Colab.

Check Version

```
import sklearn  
print(sklearn.__version__)
```

(Optional) Install PDPBox

```
!pip install pdpbox
```

Import PDP & ICE (scikit-learn)

```
from sklearn.inspection import PartialDependenceDisplay
```

- Supports both **PDP and ICE** via `kind="average"` or `kind="individual"`



Import PFI (Permutation Feature Importance)

```
from sklearn.inspection import permutation_importance
```

Note: Restart runtime after installation changes.